

YUSHI WANG

PRINCIPAL EXPERIENCE DESIGNER · DESIGN TECHNOLOGIST — AEC · BIM · AI · FULL-STACK

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PROFILE

Background	Sole design technologist at a 200-person global AEC firm, turning workflow bottlenecks and pain points into solutions that advance design clarity, performance and collaboration.
Experience Design	Runs the full design arc for technical AEC and BIM tools: research into product improvements, low-to-high-fidelity prototypes, and validation with the designers, Senior Associates and PMs who use them.
Impact	Over a decade in the Autodesk AEC stack (Revit, AutoCAD, Dynamo) and real BIM workflows. Built an AIA TAP Award-winning sustainability platform used in 85+ countries and an AI rendering pipeline that cut output ~70% and \$50K+ outsourcing.
Enablement	Founded and led an internal AI research group and a 100+ member Computational Design Community, turning scripts into structured teaching and knowledge-capturing AI tools across AIGC, parametric design and marketing.
Profile	Goal-oriented and detail-minded. Cuts to the core of a messy problem, names the real bottleneck, and pushes the fix to release. Uses vibe coding to turn ideas into validated prototypes within days. Communicates clearly in writing, in person and in visuals; fluent English; fast to learn.

SELECTED PROJECTS

Interactive Virtual Mockup (IVM)

2024 — 2025

Experience Design · Immersive VR / AR

- Role** Physical healthcare mockups cost \$12,000–\$18,000 per room and cannot show movement or workflow. Built a multi-user VR tool where surgeons, nurses, infection-control staff and architects walk a clinical room before construction, operate equipment, switch options and give structured feedback in context.
- Build** Pipeline from Rhino/Revit into Unity and Meta Quest, shipped over three releases: an interactive base with screenshot and voice annotation, button-driven equipment controls under higher-fidelity lighting, then synchronized multiplayer review of multi-layer ceilings; an AR track places full-scale options on the real site.
- Impact** Saved \$112,000 across seven Dana-Farber mockup rooms and cut medical-gas change orders 60% by catching clearance and workflow conflicts before ground-breaking. Adopted at White Plains Hospital and Dana-Farber, where the Chief of Surgery called it a major advancement in surgical facility design.

Floorcast

Feb 2026 — Present

Experience Design · AI Building-Use Simulation

- Role** Predicts how a building will really be used before it is built: AI agents play each clinical role to reproduce real routes and workflows and expose how a layout performs.
- Build** Movement, comfort and acoustic heatmaps with live data give the quantitative picture; an AI interview system adds the qualitative why. Extending from one-click 2D-to-3D to a collaborative Revit/Rhino plugin that runs simulation inside the design workflow.
- Impact** Analyzed nurse call-response and rounding routes to test nurse-station placement, then re-strategized the equipment and medication room doors to shorten clinical circulation.

PAYETTE Places

Mar — Jun 2025

Experience Design · Interactive Web

- Role** Designed the navigation and wayfinding experience for the firm public project tour at the 2025 AIA Conference, serving architects, clients and press across eight built Boston projects.
- Build** Information architecture spanning an interactive city map with one-tap Google Maps routing, building hotspots that open project pages, a navigable gallery, and animated facade diagrams with walking figures; fully responsive, with the site QR on company business cards.
- Impact** Became the public-facing showcase for AIA 2025 and a benchmark for how the office presents its work, with roughly 300 visitors on a single open day.

Promptitect

Sep 2025 — Present

Product & Experience Design · AI Prompt Intelligence

- Role** An AI prompt-intelligence layer that turns scattered, one-off AI experiments into a shared, continuously improving capability.
- Build** A searchable knowledge gallery, RAG case retrieval with match scores, a guided prompt builder, and a node-based board for comparing model outputs. Tuned so someone with no AI background gets reliable results in fewer steps, with copyright and IP compliance built in.
- Impact** Standardized prompt quality across AIGC, parametric design and marketing while protecting copyright and lowering the barrier for everyday designers.

PROFESSIONAL EXPERIENCE

PAYETTE — Boston, MA

June 2022 — Present

Design Technologist

- Validation & Feedback** Built the client-feedback loop before construction: Pulse, a web tool for structured feedback on 2D plans and digital design options, plus immersive AR/VR review in Unity, C# and iOS ARKit for clinical, engineering and architecture teams to validate space and equipment on site.
- User Research & Experience** Own user research and interaction design for internal tools serving 200+ architects, engineers and BIM staff: space-strategy (Floorish, Sectioneer, Floorcast), sustainability (Kaleidoscope, an embodied-carbon platform that won the AIA TAP Award across 85+ countries) and the IVM healthcare VR tool. Run interviews, workflow analysis and usability testing.
- Computational Design** Build professional AEC tooling hands-on: 30+ Grasshopper and Dynamo scripts, a .NET Revit add-in installing a company plugin ribbon across the office, Python Rhino plugins and generative design; consolidated into the firm script library, training and regular AI lectures.
- Applied AI** Drove applied AI end to end: led the AI group that tuned image-to-image generation into a ~70%-faster rendering pipeline (\$50K+ outsourcing saved), built a prompt-intelligence tool and an AI video tool for project-bid storytelling, and use vibe coding, AI agents and RAG to ship internal web tools.
- Full-Stack Delivery** Take 10+ tools from concept to production largely single-handed: meet with the three practice Principals and project-group PMs to surface needs, build front and back end, deploy on Azure with versioned releases, and land specs with engineering, lifting workflows 50–80%. Also built the Ask Tom RAG agent and Payette Lens (AI financial-data visualization).
- Leadership & Management** Founded and lead the firm AI research group and a 100+ member design-tech community: set multi-year tooling and AI strategy, build training, onboarding and version-controlled knowledge systems, govern design and IP standards, mentor designers across offices, and report monthly to firm Principals and leadership.

Dowbuilt — Seattle, WA

Dec 2021 — June 2022

Digital Construction Project Engineer

- BIM Automation** Built a Python/Revit BIM modeling and automated quantity-take-off system; one-click material lists and cost estimates cut take-off from days to minutes (about 90% faster), with real-time cost data across 10+ high-end residential projects.
- Pre-construction & Field** Delivered Grasshopper/Dynamo/Python tooling for excavation modeling, 4D scheduling, clash detection and CNC-ready fabrication; coordinated weekly with PMs, superintendents and fabricators.

INTERNSHIP

NBBJ — Seattle, WA

July — Sept 2021

Digital Design Specialist Intern

- UI/UX** Led a full UI/UX redesign of the firm enterprise digital-technology platform (unified tool search and technical-asset management), improving usability and cross-office collaboration.
- Parametric R&D** Built parametric Grasshopper/Dynamo tools (brick-porosity, solar-tracking façade, automated daylight simulation) adopted in pilot projects and QA/QC.

EDUCATION

University of Southern California	M.S. Building Science · GPA 4.0 / 4.0	Los Angeles · 2019 — 2021
Shanghai University of Engineering Science	B.S. Construction Management · GPA 4.0 / 4.0	Shanghai · 2016 — 2018

SKILLS & CERTIFICATE

SKILLS User Research (qual + quant) · Usability Testing · Interaction Design · Prototyping · Workflow Analysis · Data-Driven Design · Cross-functional Influence

DESIGN Figma · Adobe Creative Cloud · InDesign · Claude Design

COMPUTATIONAL DESIGN Rhino · Grasshopper · Revit · Dynamo · Rhino.Inside

DEVELOPMENT JavaScript · HTML/CSS · React · Node.js · Python · C# · Azure · Unity · iOS ARKit · MySQL · Power BI · Tableau · D3

AI Claude Code · Codex · Gemini · ComfyUI · xfigura · Krea · Runway · Kohya · RAG

AWARDS & CERTIFICATE LEED AP (USGBC, 2021) · 2022 ARCHITECT R+D & AIA TAP Innovation Award (Kaleidoscope) · IESLA Russell Cole Design Competition, Advanced (2020)